

Math 2263 Midterm 1 Review

Topics

1. Level curves
2. Limits (including switching into polar and showing limits do not exist)
3. Partial derivatives and the chain rule (using the tree-like diagrams)
4. Directional derivatives and the gradient vector:
 - (a) If $\mathbf{u}$ is a unit vector, $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$.
 - (b) The maximum possible directional derivative is $|\nabla f|$, found when $\mathbf{u} = \nabla f / |\nabla f|$
 - (c) ∇f is perpendicular to level curves
5. Finding tangent planes and normal lines
6. Finding absolute maximums and minimums for $f(x, y)$ over a region R
7. The second derivative test to identify local maximums, local minimums and saddle points
8. Constrained optimization and Lagrange multipliers

Sample questions

1. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^2 + y^2}$ or show this limit does not exist.
2. Suppose $ze^{xyz} = 0$ where z is an unknown function of x and y . Write z_x in terms of x, y , and z .
3. Find the directional derivative of $f(x, y) = 3x^2 + 2xy$ in the direction of $\langle 2, 3 \rangle$ at $(1, 1)$. What does this calculation mean? (Explain with words.)
4. Find and classify all local minimums and maximums for $x^3 - 6xy + 8y^3$.
5. Find the maximum and minimum for the function $(x - 1)^2 + (2y - 1)^2$ on the triangle with corners $(0, 0)$, $(2, 0)$, and $(0, 2)$.
6. Let S be the set of points which satisfy $z - e^{x+xy} = 0$ and let p be the point $(1, -1, 1)$. Find the plane tangent to S at p and find the line normal to S at p .
7. Let $z = f(x, y)$ with $x = \cos(st)$ and $y = s - t$. Find $\frac{\partial^2 z}{\partial t^2}$ in terms of $z_{xx}, z_{xy}, z_{yy}, s$ and t .
8. Find the point on the sphere $x^2 + y^2 + z^2 = 1$ which maximizes $x + 2y + z$.
9. Minimize $x^2 + 2xy + y^2$ subject to the constraint $x^2 + y^2 = 1$.